

Burton Rasmussen

Mechanical Engineer | Mechatronics Emphasis

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Project Portfolio: burtonrasmussen.github.io

EDUCATION

University of Utah | *College of Engineering*

Anticipated Graduation: Spring 2027

B.S. Mechanical Engineering | Mechatronics Emphasis • GPA: 3.4

SKILLS

- **CAD & Mechanical Design:** SOLIDWORKS (CSWA - Certified SOLIDWORKS Associate), OnShape, Fusion 360, SOLIDWORKS PDM, Design for Manufacturability (DFM), GD&T, Sheet Metal Modeling, Finite Element Analysis (FEA)
- **Quality Systems & Validations:** Process Validation (IQ/PQ - Installation & Performance Qualification), Measurement System Analysis (MSA), Standard Operating Procedures (SOP/SWI), Change Controls
- **Coding and Data Analysis:** Python (Statistical Process Control / p-charts), MATLAB, Database Analysis, Data Acquisition (DAQ) Systems
- **Manufacturing & Prototyping:** Manual & CNC Milling, 3D Printing (FDM/SLA), Fixtures & Inspection Tooling
- **Mechatronics:** Embedded Systems, Bare Metal C programming, C++ (Arduino), Systems & Controls, Programmable Logic Controllers (PLCs), KiCAD PCB Design, Soldering, Wire Harnesses
- **Languages & Certifications:** Spanish (B2 Proficiency Level), Certified IPC Specialist (IPC/WHMA-A-620)

EXPERIENCE

Becton Dickinson (BD) | *Process Engineering Intern*

Feb 2026 – Present

- Diagnosed fixture damage and alignment by developing a Python SPC tool to isolate statistical outliers in production data, communicating targeted daily maintenance tasks with maintenance technicians that reduced scrap by 25%.
- Mentored and trained 2 incoming interns, while assisting line operators and maintenance with daily root cause analysis and equipment troubleshooting.
- Identified rotary dial bottlenecks using cycle rate data to find tooling and PLC timing optimizations to cut 200 ms per cycle and increase capacity from 90 to 110 parts/min (+22%).
- Reduced jams in feeder bowl escapement tooling by revising tooling geometry based on design review feedback and applying GD&T principles to reduce machine downtime by 40%.
- Authored IQ/PQ Validation Protocols and navigated formal Change Controls for upgraded adhesive dispensing equipment as part of a Kaizen event.

University of Utah | *Undergrad Researcher*

Summer – Fall 2025

- Developed an automated experiment setup utilizing an RS-485 PID heater controller, custom servo-actuated valves, and DAQ hardware, using MATLAB to corroborate sensor and Arduino microcontroller data.

ATL Technology | *Manufacturing Engineering Technician*

April 2023 – July 2024

- Improved assembly and inspection throughput by designing fixture in SOLIDWORKS to aid in alignment and visual verification that product meets specification in one step, saving \$7,000 in annual labor and scrap costs.
- Conducted Measurement System Analysis to validate challenge samples for electrical quality control test equipment.
- Authored SOPs & validation documentation via engineering Change Controls to support manufacturing operations.

PROJECTS

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Computerized Telescope (UVU E2i Grant Recipient)

- Designed a portable truss structure in OnShape using FEA, sheet metal modeling, and 3D Printing DFM principles.
- Fabricated parts utilizing 3D printing and CNC sheet cutting to achieve rigidity and precision for optical alignment.

Mechatronics Robot (2nd Place Competition)

- Developed KiCad schematics, executing soldering and wire harness assembly.
- Implemented omni-wheel trajectory control in C++, providing a winning navigational advantage.

Inverted Pendulum Balancing Robot

- Designed a motor driver PCB and schematic in KiCad to interface with encoders and control the cart motor via an STM32 for an Embedded Systems final project.
- Implemented an LQR state feedback control model with a 1kHz loop, successfully handling external disturbances while maintaining balance and returning to position.